#In this code we will demonstrate the Proposition 7.23.

#Define f1 and f2

$f1 := w^{-3} + z^{-3} - 2 \cdot t^{-3};$

$f2 := (w^{-3} - x^{-3}) \cdot (z^{-3} - w^{-3}) + (x^{-3} - t^{-3}) \cdot (y^{-3} - w^{-3});$

$$f1 := \frac{1}{w^3} + \frac{1}{z^3} - \frac{2}{t^3}$$

$$f2 := \left(\frac{1}{w^3} - \frac{1}{x^3}\right)\left(\frac{1}{z^3} - \frac{1}{w^3}\right) + \left(\frac{1}{x^3} - \frac{1}{t^3}\right)\left(\frac{1}{y^3} - \frac{1}{w^3}\right) \tag{1}$$

#By construction, we have the following relations

$t := s;$
$s := \frac{y}{2};$
$x := \frac{w^2 - z^2}{y};$

$$t := s$$

$$s := \frac{y}{2}$$

$$x := \frac{w^2 - z^2}{y} \tag{2}$$

#Performing the following rational parametrization

$z := y \cdot \frac{2 \cdot \xi}{1 + \xi^2}; w := y \cdot \frac{1 - \xi^2}{1 + \xi^2}$

$$z := \frac{2 y \xi}{\xi^2 + 1}$$

$$w := \frac{y\left(-\xi^2 + 1\right)}{\xi^2 + 1} \tag{3}$$

#Define g1

$g1 := numer(f1)$

$$g1 := \xi^{12} - 136\,\xi^9 - 3\,\xi^8 + 360\,\xi^7 - 408\,\xi^5 + 3\,\xi^4 + 120\,\xi^3 - 1 \tag{4}$$

#Define g2

$g2 := numer(f2)$

$$g2 := -y^3\,\xi^2\,\left(\xi^{28} - 6\,\xi^{26} + 64\,\xi^{25} + 27\,\xi^{24} - 1392\,\xi^{23} + 28\,\xi^{22} + 11040\,\xi^{21} - 255\,\xi^{20} - 44336\,\xi^{19}\right. \tag{5}$$

$$- 210\,\xi^{18} + 109824\,\xi^{17} + 611\,\xi^{16} - 205536\,\xi^{15} + 552\,\xi^{14} + 267200\,\xi^{13} - 597\,\xi^{12} - 249312\,\xi^{11} - 586\,\xi^{10} + 123840\,\xi^{9} + 249\,\xi^{8} - 31664\,\xi^{7} + 252\,\xi^{6} + 4128\,\xi^{5} - 45\,\xi^{4} - 240\,\xi^{3} - 30\,\xi^{2} + 9\Big)$$

#Define G2

$$G2 := \xi^{28} - 6\,\xi^{26} + 64\,\xi^{25} + 27\,\xi^{24} - 1392\,\xi^{23} + 28\,\xi^{22} + 11040\,\xi^{21} - 255\,\xi^{20} - 44336\,\xi^{19} - 210\,\xi^{18} + 109824\,\xi^{17} + 611\,\xi^{16} - 205536\,\xi^{15} + 552\,\xi^{14} + 267200\,\xi^{13} - 597\,\xi^{12} - 249312\,\xi^{11} - 586\,\xi^{10} + 123840\,\xi^{9} + 249\,\xi^{8} - 31664\,\xi^{7} + 252\,\xi^{6} + 4128\,\xi^{5} - 45\,\xi^{4} - 240\,\xi^{3} - 30\,\xi^{2} + 9 :$$

#Compute the resultant

$resultant(g1, G2, \xi)$

$$-38128759262538190451210055401120006044987824285079771263729989293838350559237308940288 \qquad \textbf{(6)}$$